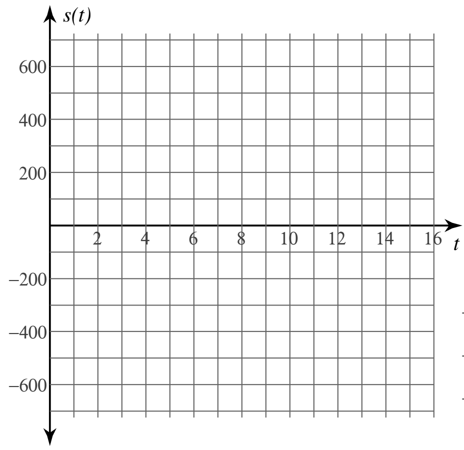
Kuta Software - Infinite Calculus   Name____________________

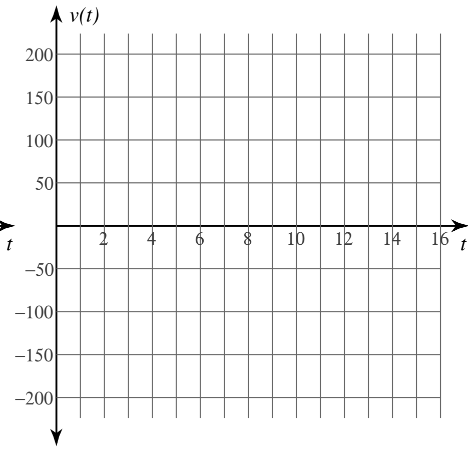
Motion Along a Line

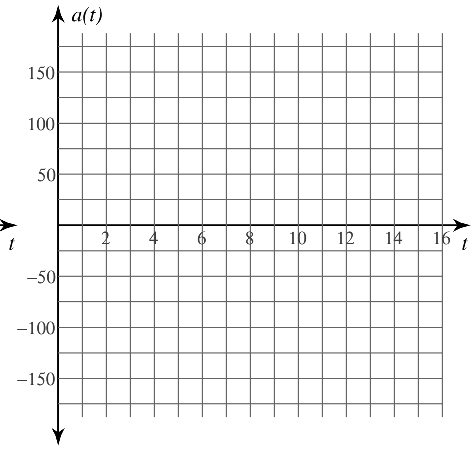
Date________________ Period____

A particle moves along a horizontal line. Its position function is $s(t)$ for $t \geq 0$. For each problem, find the velocity function $v(t)$ and the acceleration function $a(t)$. You may use the blank graphs to sketch $s(t)$, $v(t)$, and $a(t)$.

1) $s(t) = t^3 - t^2 - 56t$

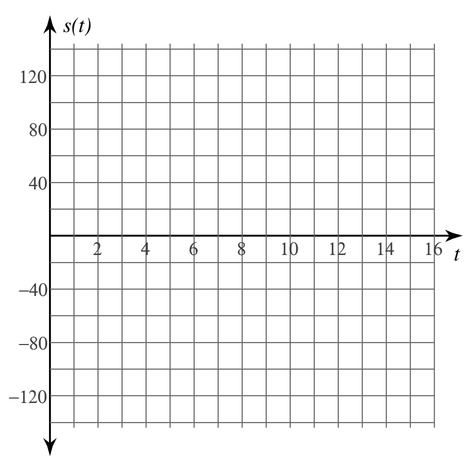






A particle moves along a horizontal line. Its position function is $s(t)$ for $t \geq 0$. For each problem, find the displacement of the particle and the distance traveled by the particle over the given interval. You may use the blank graph to sketch $s(t)$.

2) $s(t) = -t^2 + 6t + 27;\ 0 \leq t \leq 4$



3) $s(t) = -t^3 + 11t^2;\ 3 \leq t \leq 8$